

Grade 8

Unit 8

Lesson 10 Homework

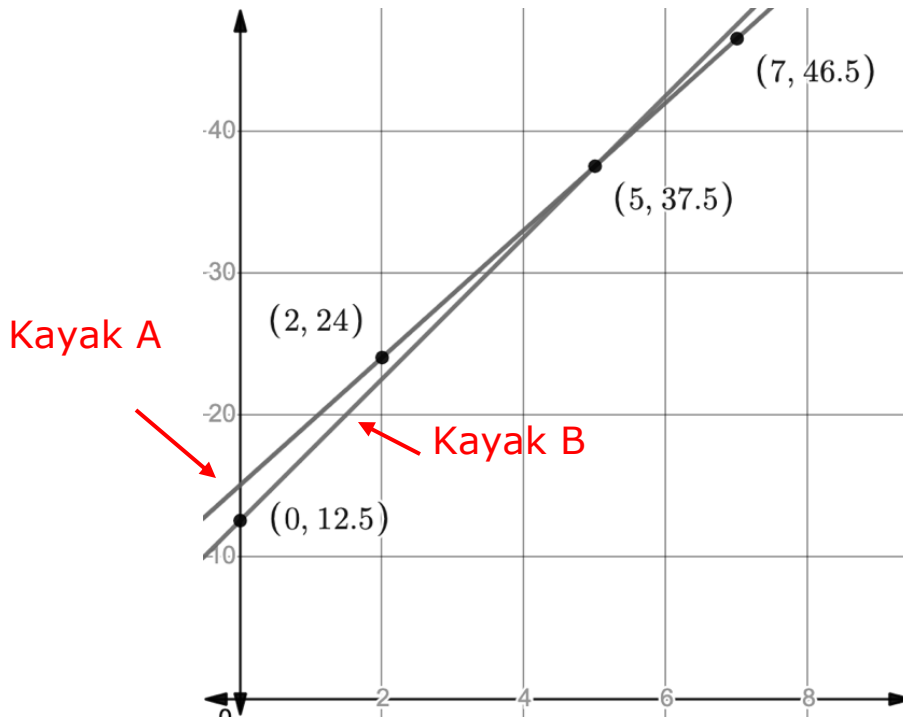
Name Answer Key

Date _____

Period _____

Noah is planning a kayaking trip. Kayak Rental A charges a base fee of \$15 plus \$4.50 per hour. Kayak Rental B charges a base fee of \$12.50 plus \$5 per hour.

The system $\begin{cases} y = 15 + 4.5x \\ y = 12.50 + 5x \end{cases}$ represents this situation, where x is the number of hours and y is the cost. The graph of the system is shown with *four* points labeled.



1. Label the line that represents Kayak A.
2. Label the line that represents Kayak B.
3. What is the solution to the system?
(5,37.5)
4. Explain the meaning of the solution in this context.
After 5 hours, the cost for both kayaks will be \$37.50.

Andre and Elena are each saving money. Andre starts with \$40 in his savings account and adds \$5 per week. Elena starts with \$10 in her savings account and adds \$20 each week.

The system $\begin{cases} y = 40 + 5x \\ y = 10 + 20x \end{cases}$ represents this situation, where x is the number of weeks and y is the amount of money saved.

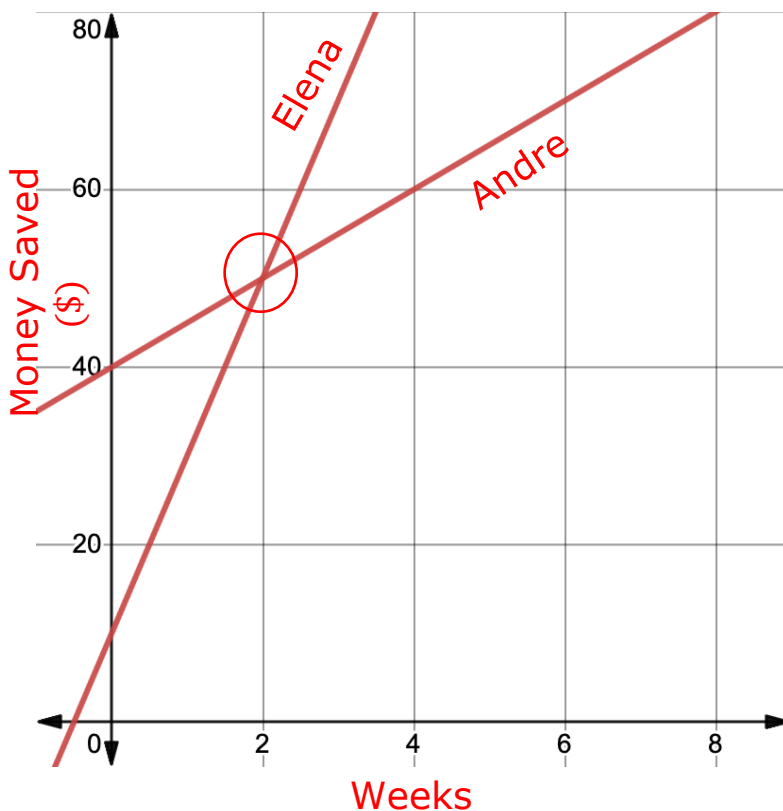
5. Graph the system.

6. Label the axes with the appropriate titles.

7. Label each line with "Andre" or "Elena".

8. What is the solution to the system?
Circle it on the graph.

(2,50)



9. Explain the meaning of the solution in the context?

After 2 weeks, both Elena and Andre will have \$50 saved.

10. Explain why this graph would not be a reasonable way to determine how much Andre and Elena have both saved after 1 year (52 weeks).

Answers may vary.

The axes are scaled too small for that, and the amount saved will be too great.